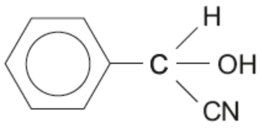
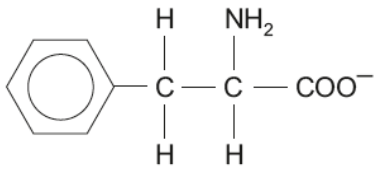
Section A

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
1.	*(a)*					1		1		
	(b)			^{-}CN accept CN^{-}	1			1		
2.				butanoic acid accept methylpropanoic acid / pentanedioic acid		1		1		
3.				2 mol NH_3 from 1 mol amide **T** 0.060 mol from 0.030 mol amide **T** M_r of the amide 3.90/0.030 = 130 (1) M_r 'R' = 130 – 88 = 42 therefore C_3H_6 / $CH_2CH_2CH_2$ (1)		1	1	2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	*(a)*					1		1		
	(b)			2-aminopropanoic acid exists as zwitterions / $CH_3CHNH_3^+COO^-$ (1) it has strong ionic character therefore much stronger forces between molecules than the other acids (1)		2		2		
5.				275.4 dm^3 gaseous material from 222 g RDX (1) therefore 1 m^3 from 222 × 1000 / 275.4 = 806 answer must be given to 3 sig figs (1)		1	1	2	2	
				Section A total	**1**	**7**	**2**	**10**	**3**	**0**